BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 234 · Spiral model and Rapid Application Development

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Name a key XP programming practice.

Q2. How does an Agile sprint help with changed requirements?

Q3. What is a prototype?

Q4. Why might risk analysis be valuable before developing an uncertain hardware-dependent system?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Spiral model — Iterative methodology driven by risk identification and evaluation.

risk analysis — Evaluate probability/impact of potential problems.

RAD — Rapid Application Development using frequent prototypes and feedback.

prototype — Early working or mock representation for evaluation.

stakeholder — Person or group affected by or using the system.

New learning

Spiral focuses on risk — Each cycle plans objectives, identifies or resolves major risks, develops/tests and reviews with stakeholders.

RAD is prototype-led — Users evaluate quickly produced models. Frequent prototyping accelerates understanding and can reveal unclear requirements.

Compare suitability — Spiral can suit large/high-risk systems but increases management cost; RAD relies on accessible users and skilled teams and may compromise architecture if rushed.

Worked example

A satellite control project has major safety and technology uncertainties. Recommend a method.

1. Predict first — Spiral is appropriate because major technical and safety risks need assessment at each cycle.

2. Model and explain — It demands experienced risk assessment and carries greater time/management overhead than simpler methods.

We Do · Practise with guidance

A café needs a quick booking interface but cannot clearly describe its desired layout. Recommend a method.

```
Sketch UI → build prototype → café reviews
→ revise → pilot working version
```

Predict first

RAD suits evolving interface requirements because the café can see and comment on working prototypes.

Model and explain

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

A risk is that repeated quick changes may weaken maintainability without time for robust design.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

State one key characteristic and one disadvantage of both Spiral and RAD.

Question 2

A transport company is building an experimental driverless control system. Compare Spiral with RAD and justify a choice.

Question 3

A booking website must launch rapidly and users are available daily. Explain three benefits and two risks of RAD.

Exit Ticket · Answer before checking

Spiral development and RAD are identical because both can repeat stages.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. For example pair programming, TDD or continuous integration.

Q2. It lets the team adjust the next iteration using feedback.

Q3. An early model used to explore and test an idea.

Q4. It can reveal technical feasibility issues before expensive implementation.

You Do pages · Independent questions

Q1. [4] Spiral is risk-driven [1], but adds assessment/management cost [1]; RAD creates prototypes with user feedback [1], but rapid changes can compromise design quality [1].

Q2. [6] Spiral explicitly assesses critical risks [1] and validates feasibility each cycle [1]; RAD rapidly gathers interface feedback [1], which is less central than safety risks [1]; choose Spiral [1] because controlling safety/technical uncertainty matters more than fast UI prototyping [1].

Q3. [6] Early working prototypes [1]; frequent direct feedback [1]; requirements clarified quickly [1]; helps prioritise useful features [1]; users must be available [1] and rushed design/testing can undermine quality [1].

Exit Ticket · Model answer

False. Spiral is explicitly risk-driven; RAD centres on rapid prototyping and frequent user feedback.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding